Global Knowledge®

Angular Fundamentals

Component Trees

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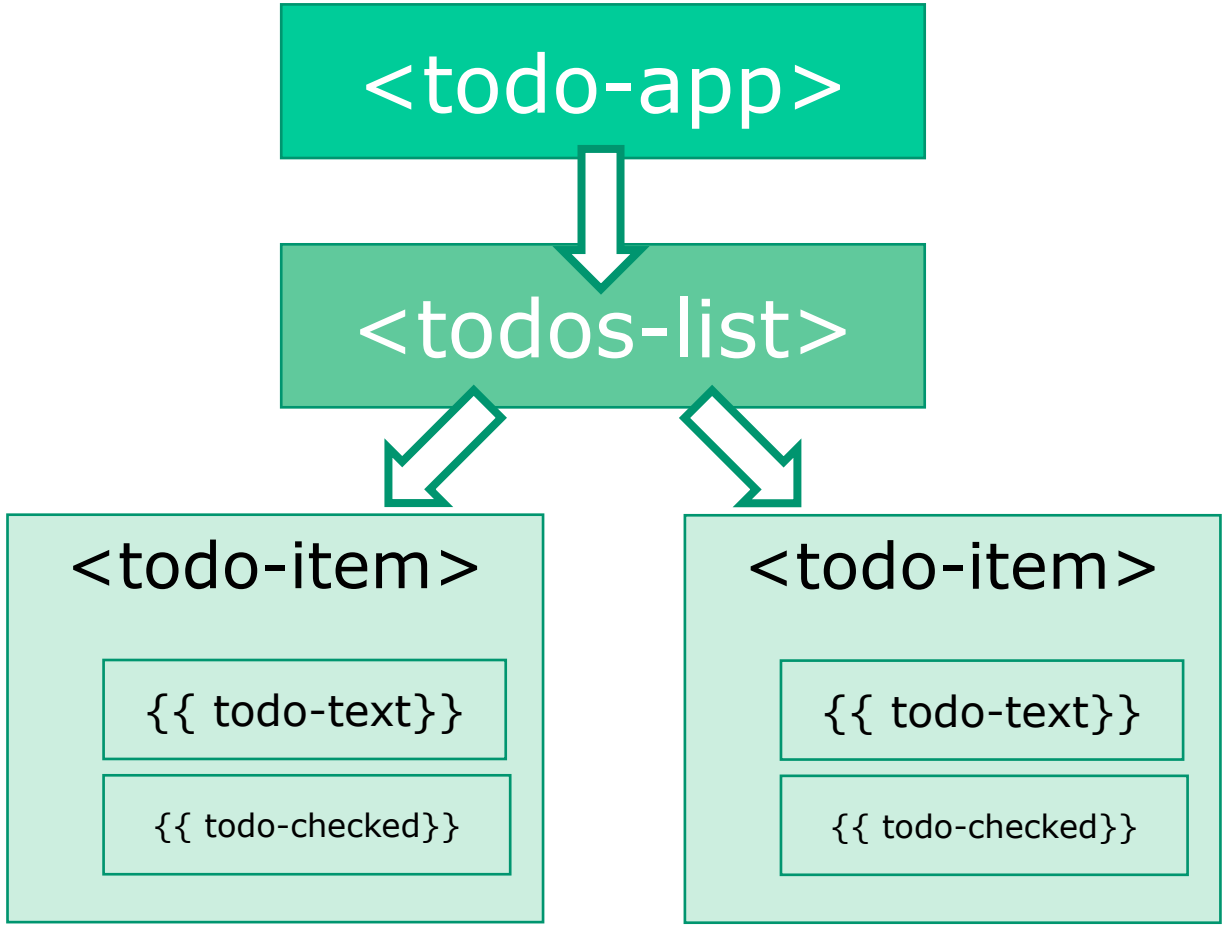
Angular-app: Tree of components



Application as a tree of components

- Recap - Multiple components?
  1. Create files manually – or let CLI handle this for you
     1. `ng generate component <component-name>`
     2. `ng g c <component-name>`
  2. Import in module – or (again) let CLI take care of this for you
  3. Add to `declarations : […]` section of `@ngModule`.
     1. IF you are working with ng modules, that is. Otherwise:

        import in component where you want to use it.
  4. Add via HTML to `parent`-component
- Repeat for every component

1. Detailcomponent toevoegen

```
// city.detail.ts
import { Component } from '@angular/core';

@Component({
   selector: 'city-detail',
   template: `
   <h2>City details</h2>
      <ul class="list-group">
         <li class="list-group-item">Naam: [naam van stad]</li>
         <li class="list-group-item">Provincie: [provincie]</li>
         <li class="list-group-item">Highlights: [highlights]</li>
      </ul>
   `
})

export class CityDetail{

}
```

Nieuwe selector

Nog in te vullen

2. Injection in Module

```
// Angular Modules
…

// Custom Components
import {AppComponent} from './app.component';
import {CityDetail} from './city.detail';
import {CityService} from "./city.service";

// Module declaration
@NgModule({
   imports      : [BrowserModule, HttpModule],
   declarations: [AppComponent, CityDetail],
   bootstrap    : [AppComponent],
   providers    : [CityService]
})
export class AppModule {
}
```

Nieuwe component

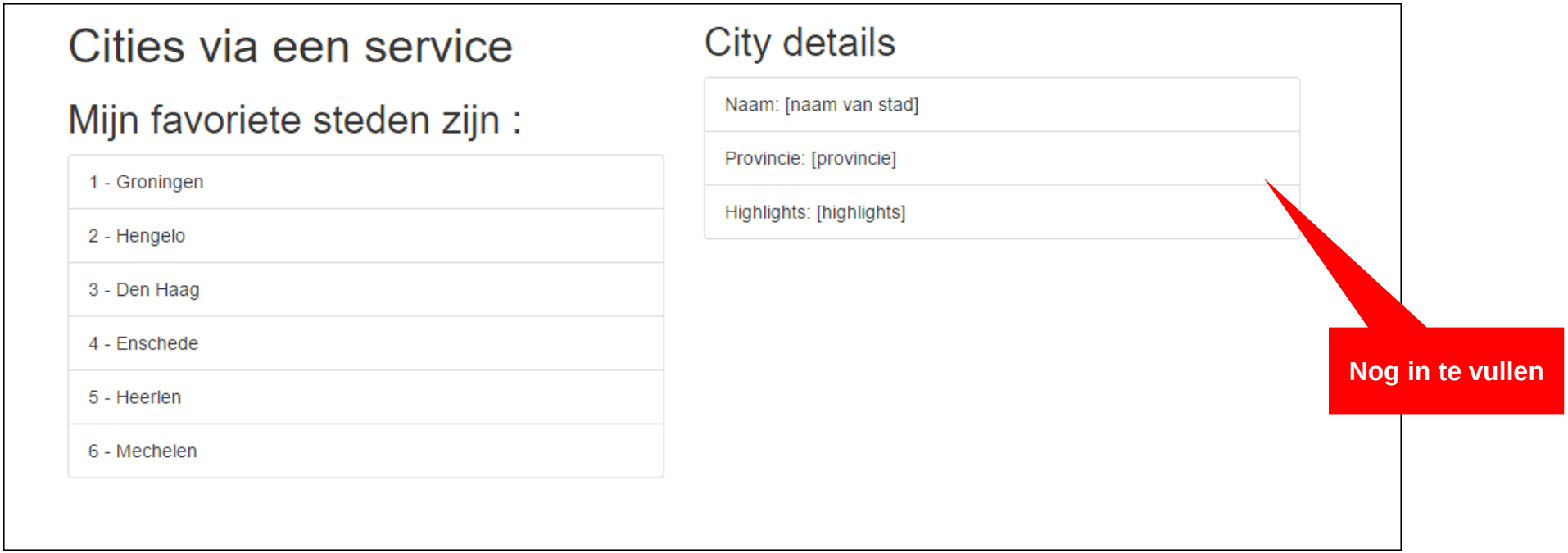
Toevoegen aan declarations: [ ]

3. Insluiten in HTML

```
<!-- app.html -->
<div class="row">
    …
   <div class="col-md-6">
      …
      <city-detail></city-detail>
   </div>
</div>
```

Combineren met overige HTML

4. Resultaat



Doel: details van geselecteerde city tonen in child-component

Data flow tussen componenten

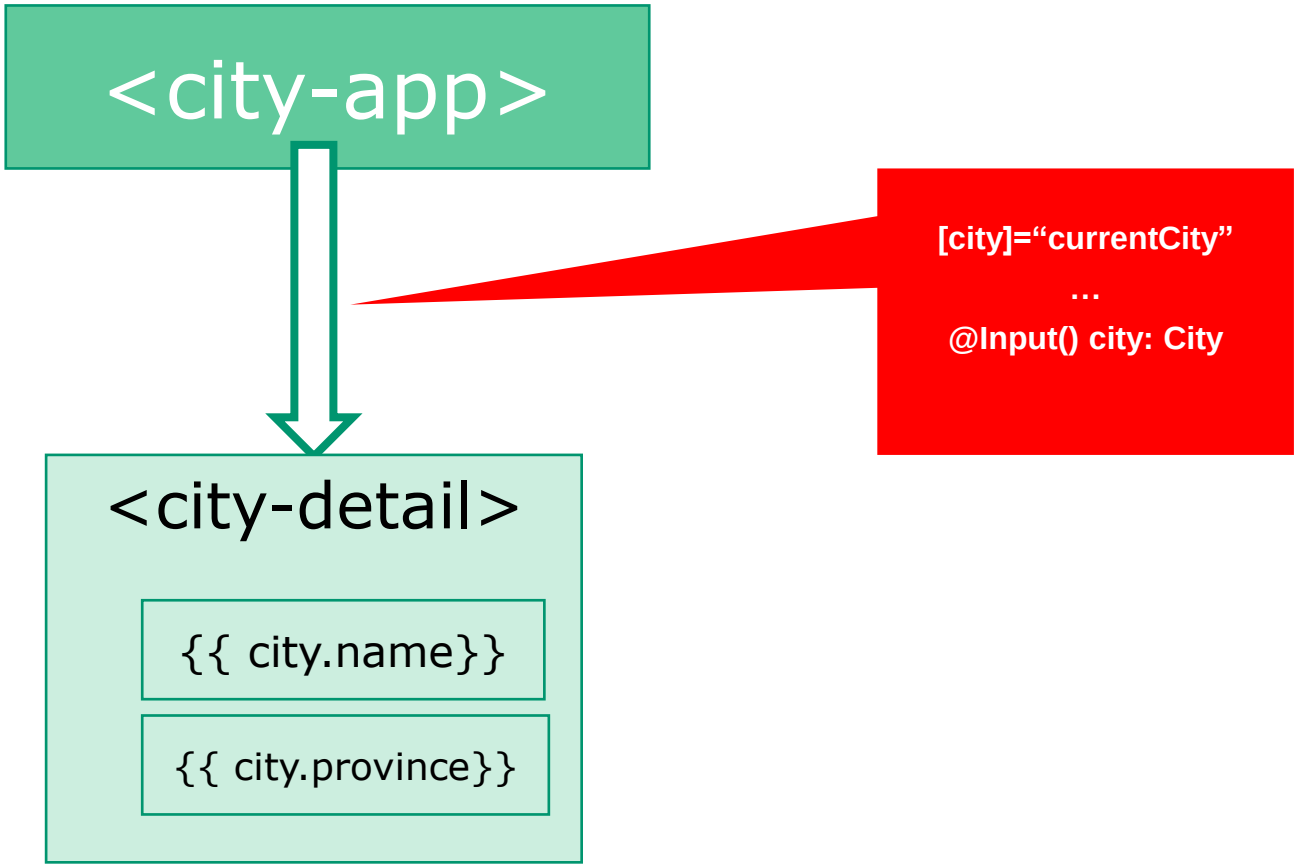
Werken met inputs en outputs

Data flow tussen components

"Data flows in to a component via `@Input()` 's"

Data flows out of a component via `@Output()`'s"

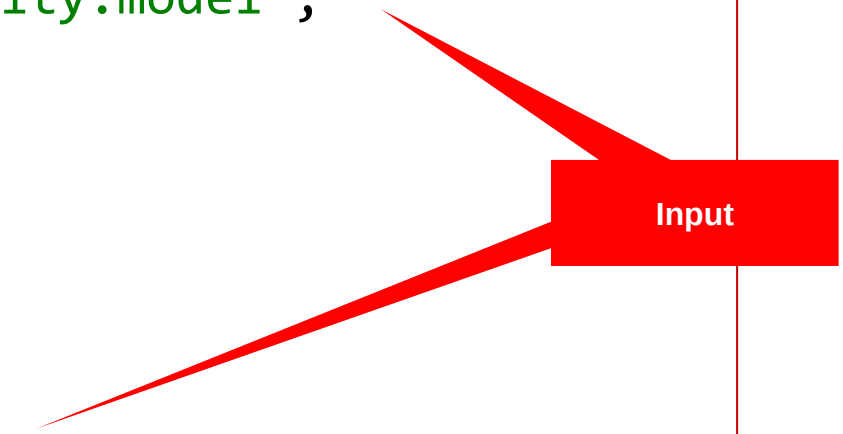
Parent-Child flow: de annotatie `@Input()`



Werken met @Input()

1. Service `Input` importeren in de betreffende component
2. Decorator `@Input()` gebruiken in de class definition

```
// city.detail.ts
import { Component, Input } from '@angular/core';
import { City } from "./city.model";

@Component({
   …
})

export class CityDetail {
   @Input() city: City;
}
```



Parent Component aanpassen voor @Input

```
<!-- app.html -->
<div class="row">
   <div class="col-md-6">
      …
      <ul class="list-group">
         <li *ngFor="“let city of cities" class="list-group-item"
                (click)="getCity(city)">
            {{ city.id}} - {{ city.name }}
         </li>
      </ul>
      <button *ngIf="currentCity" class="btn btn-primary"
                (click)="clearCity()">Clear</button>
   </div>
   <div class="col-md-6">
   <div *ngIf="currentCity">
         <city-detail [city]="currentCity"></city-detail>
      </div>
   </div>
</div>
```

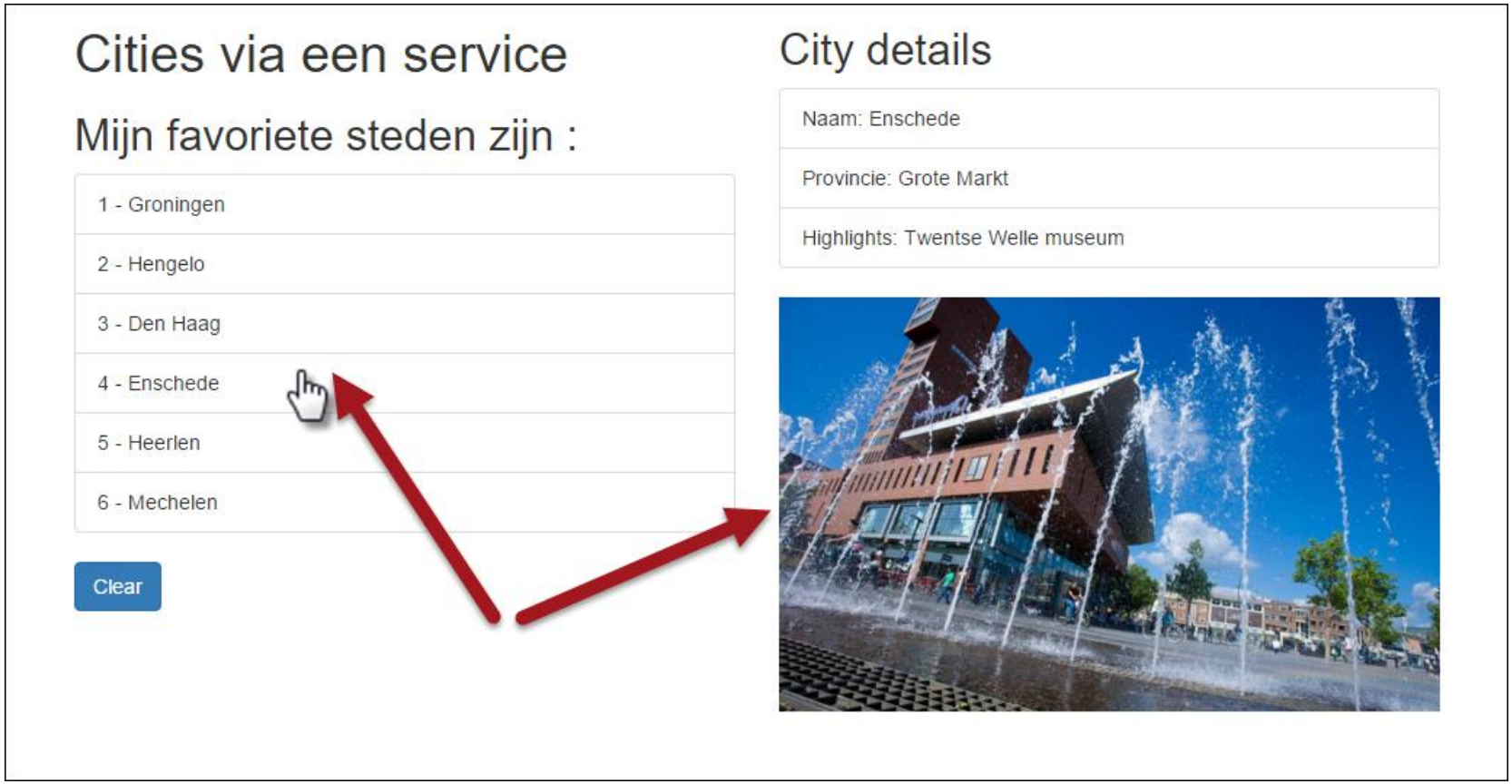
Aanpassing

Aanpassing!

Parent Component Class uitbreiden

```
export class AppComponent {
   // Properties voor de component/class
   public cities:City[];
   public currentCity:City;

   …

   getCity(city) {
      this.currentCity = city;
   }

   clearCity() {
      this.currentCity = null;
   }

   …
}
```

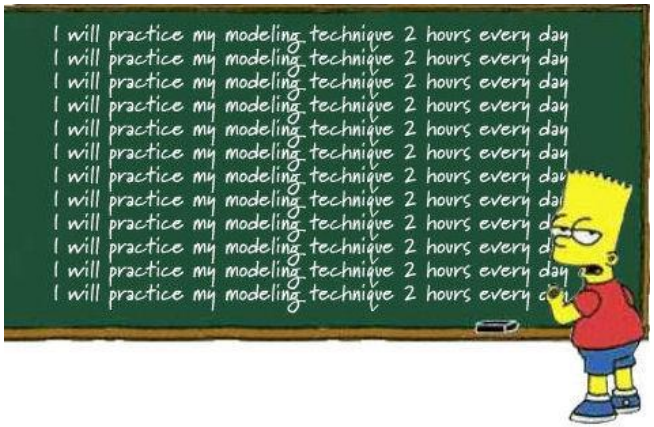
Resultaat


Cities via een service
Mijn favoriete steden zijn :
1 - Groningen
2 - Hengelo
3 - Den Haag
4 - Enschede
5 - Heerlen
6 - Mechelen
Clear
City details
Naam: Enschede
Provincie: Grote Markt
Highlights: Twentse Welle museum

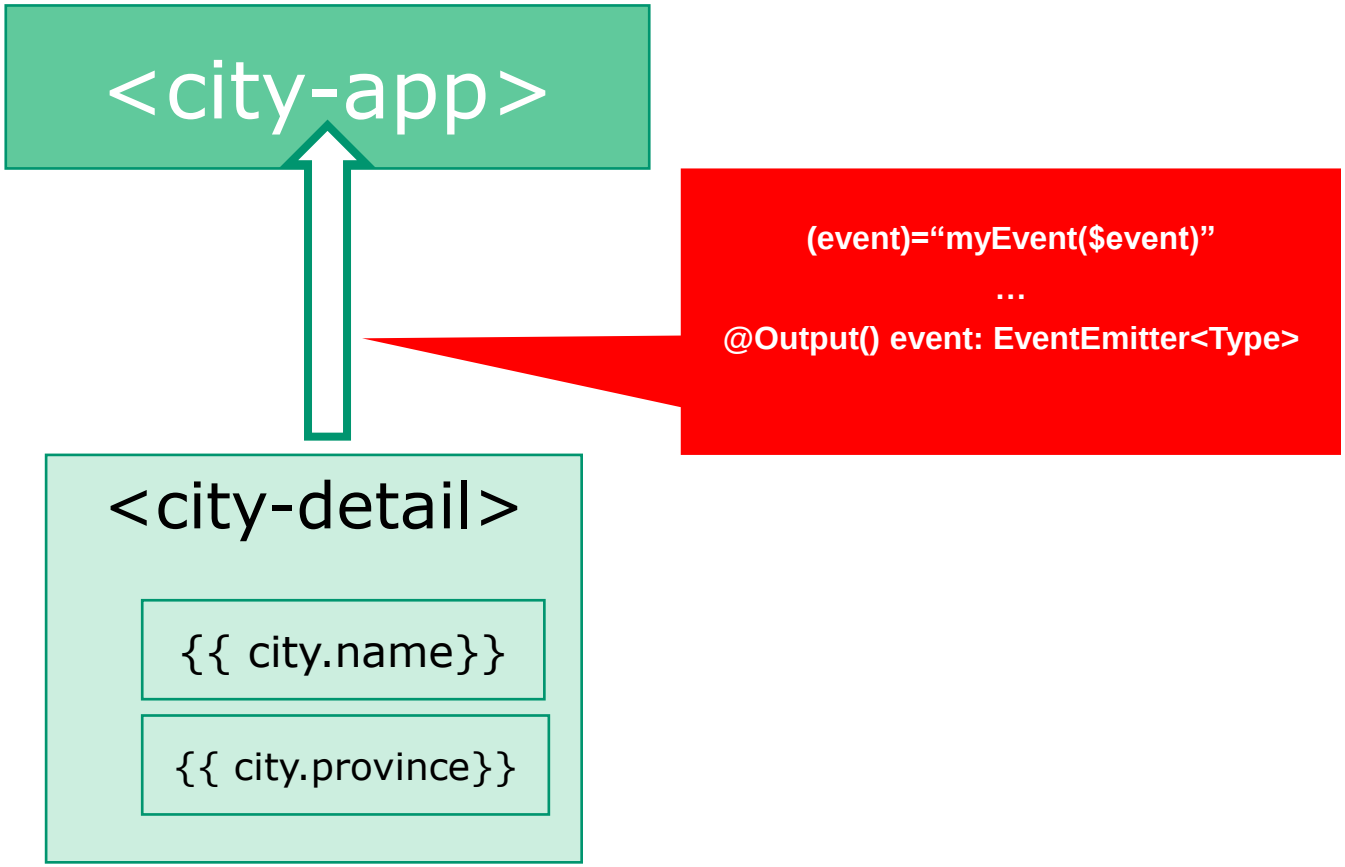
Checkpoint

- Componenten kunnen binnen componenten worden opgenomen
- Breidt de HTML van de Parent Component uit met declaratie van de Child Component
- Denk er aan Child Component te importeren in de `@ngModule`
- Data flow naar Child Component : werken met `@Input()` en `[propName]="data"`
- Oefening: `6b) en 6c)`
- Voorbeeld: `/300-components`

Oefening….



Child-Parent flow: de annotatie `@Output()`



Werkwijze – idem, maar dan andersom

1. Functie `Output` importeren in de betreffende component
2. Decorator `@Output()` gebruiken in de class definition
3. `EventEmitter` definiëren en zijn Type Annotation

“With @Output,

data flows up the Component Chain”

Een rating geven aan Cities

```
// city.detail.ts
import { Component, Input, Output, EventEmitter} from ‘@angular/core';

@Component({
   …
   template: `
   <h2>City details
      <button (click)="rate(1)">+1</button>
      <button (click)="rate(-1)">-1</button>
   </h2>
   …
   `
})

export class CityDetail {
   @Input() city:City;
   @Output() rating: EventEmitter<number> = new EventEmitter<number>();

   rate(num) {
      console.log('rating voor ', this.city.name, ': ', num);
      this.rating.emit(num);
   }
}
```

Imports

Bind custom events to DOM

Define & handle custom @Output event

Parent Component voorbereiden op ontvangen custom event

```
<!-- app.html -->
<div *ngIf="currentCity">
   <city-detail [city]="currentCity" (rating)="updateRating($event)">
   </city-detail>
</div>
```

Capture event

```
// app.component.ts
// increase or decrease rating on Event Emitted
updateRating(rating){
   this.currentCity.rating += rating;
}
```

Rating tonen in HTML

```
<li *ngFor="“let city of cities"
    class="list-group-item" (click)="getCity(city)">
   {{ city.id}} - {{ city.name }} ({{i}})
   <span class="badge">{{city.rating}}</span>
</li>
```

Rating

Resultaat

Cities via een service

Mijn favoriete steden zijn :

1 - Groningen	0
2 - Hengelo	0
3 - Den Haag	-3
4 - Enschede	0
5 - Heerlen	2
6 - Mechelen	5

Clear

City details +1 -1

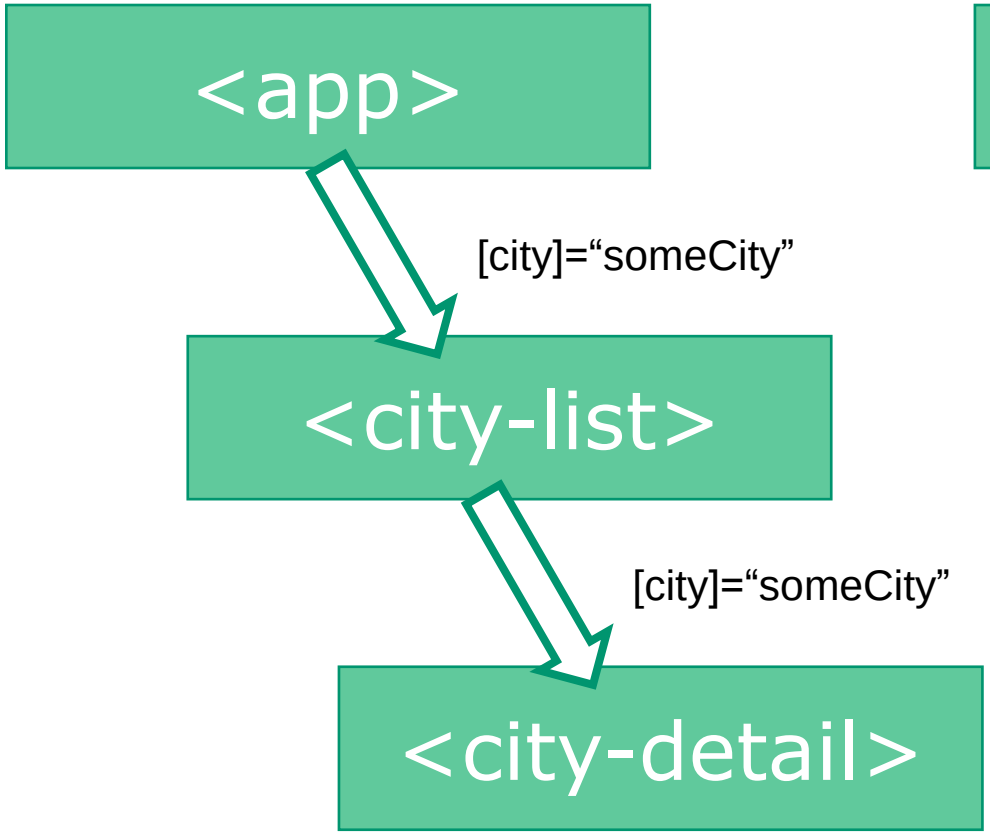
Naam: Den Haag

Provincie: Zuid-Holland

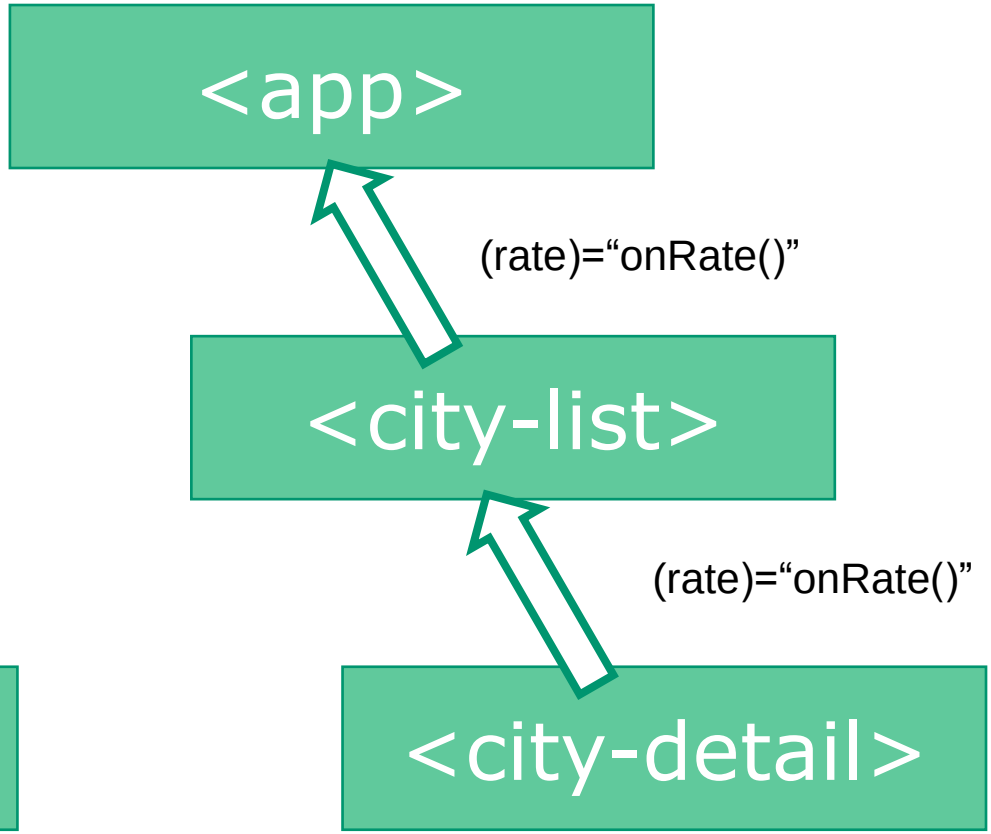
Highlights: Binnenhof

Samenvatting

Parent → Child



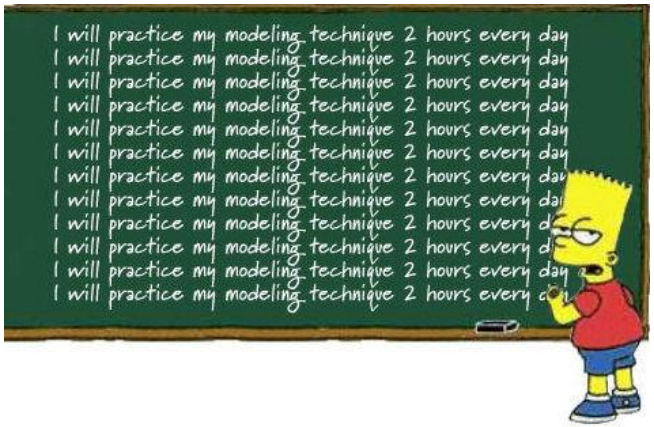
Child → Parent



Checkpoint

- Data flow naar Parent Component : werken met `@Output()` en `(eventName)="eventHandler($event)"`
- Je kunt *elk type* data uitsturen via de `EventEmitter`.
- Oefening: `6d)`
- Voorbeeld: `/302-components-output`
- Meer info: https://vsavkin.com/the-core-concepts-of-angular-2-c3d6cbe04d04

Oefening….

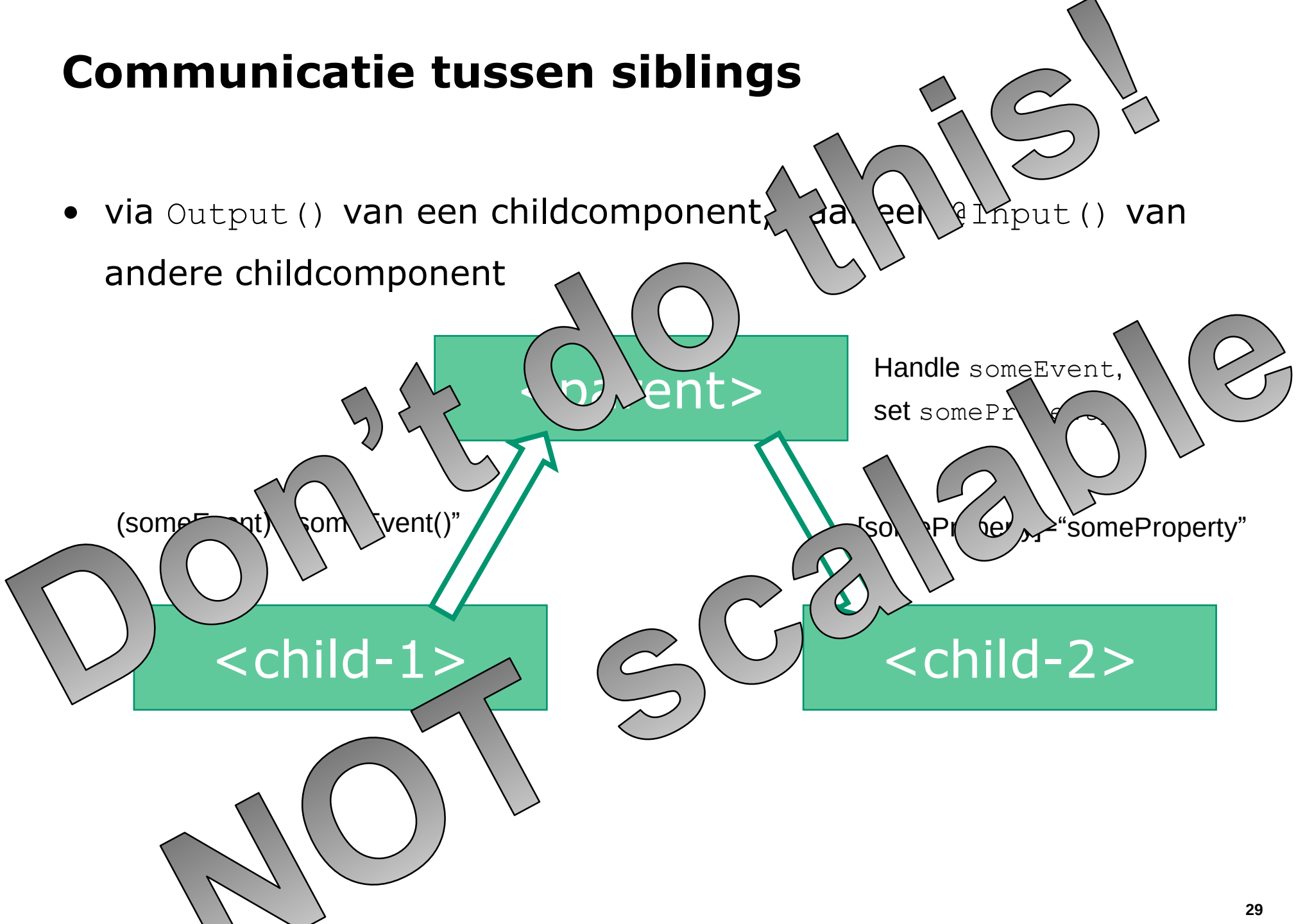


Sibling communication

Geen directe parent-child relatie tussen componenten

Communicatie tussen siblings

- via `Output()` van een childcomponent, daar een `@Input()` van andere childcomponent



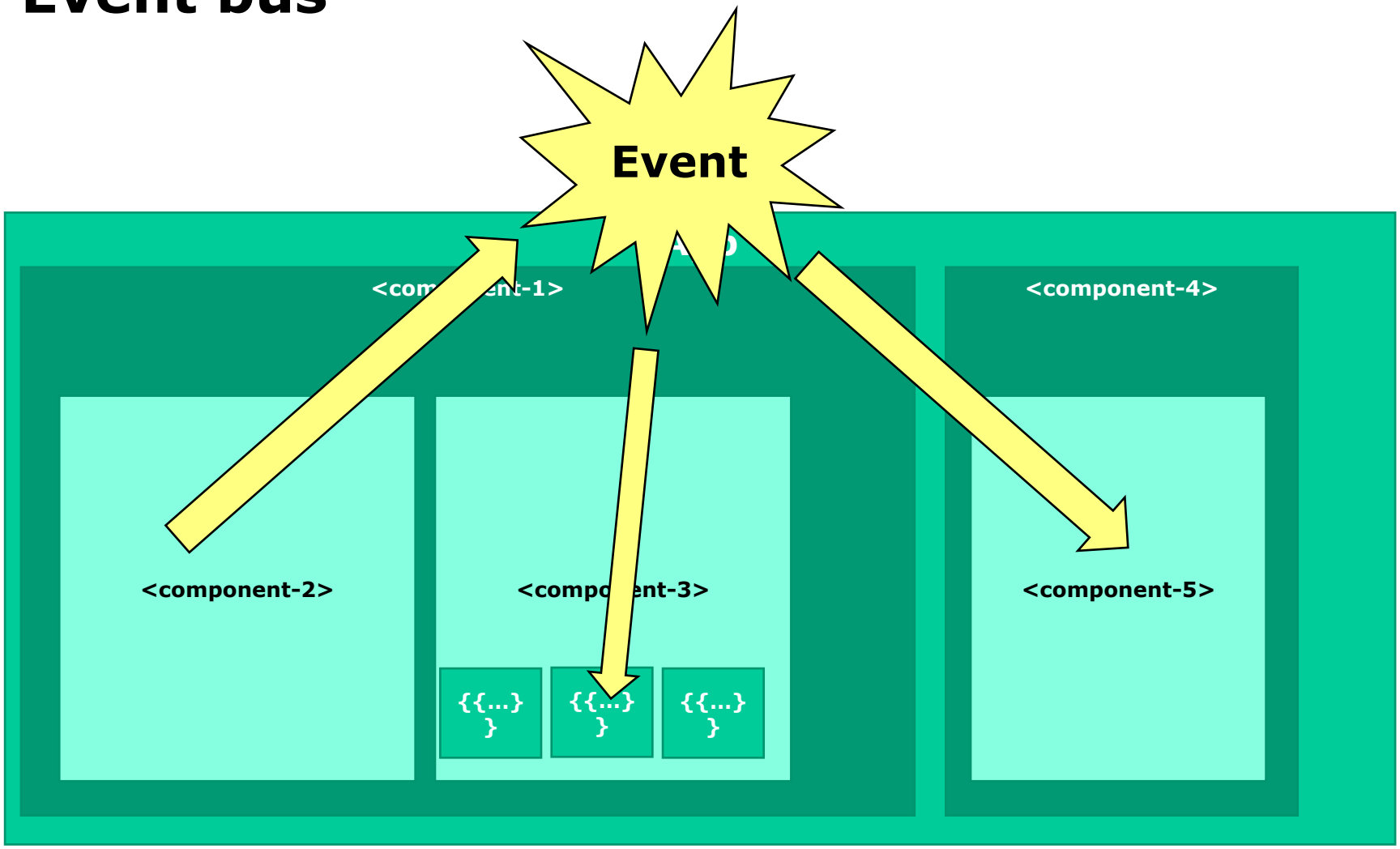
Mooiere oplossing – Pub/Sub-systeem met Observables

- http://www.syntaxsuccess.com/viewarticle/pub-sub-in-angular-2.0

“Custom events,

gebruik een event bus”

Event bus

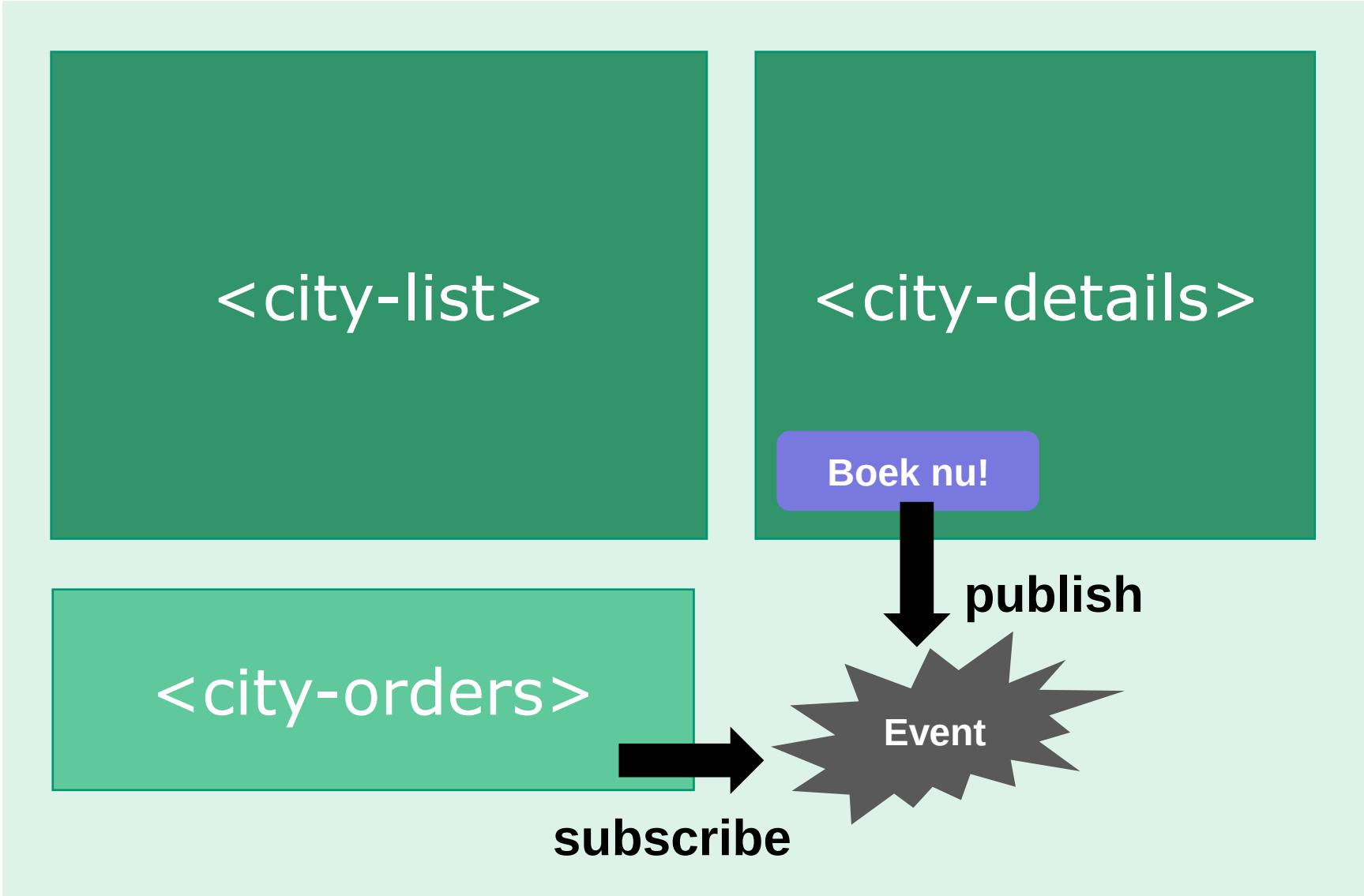

Event
<component-1>
<component-2>
<component-3>
{{...}
}
{{...}
}
{{...}
}
<component-4>
<component-5>

Opties

Uit RxJs-bibliotheek, werken met:

- `EventEmitter()`
- `Observable()`
- `Observer()`
- `Subject()` (zowel `Observable` als `Observer`)

"*Publish en Subscribe*" – PubSub systeem


<city-list>
<city-details>
Boek nu!
publish
<city-orders>
Event
subscribe

PubSub-service maken

- Stap 1 – Publicatie service maken
- Stap 2 – ‘Producer’, of ‘Publish’ – component maken
- Stap 3 – subscriber-component maken, of toevoegen aan bestaande component.

1. OrderService

```
// order.service.ts
import {Subject} from "rxjs/Subject";
import {Injectable} from "@angular/core";
import {City} from "../model/city.model";

@Injectable()
export class OrderService {
   Stream:Subject<City>;

   constructor() {
      this.Stream = new Subject<City>();
   }
}
```

2. Producer component (‘boek nu’-knop)

In de HTML:

```
<h2>Prijs voor een weekendje weg:
{{ city.price | currency:'EUR':true:'1.2' }}
<button class="btn btn-lg btn-info"
   (click)="order(city)">Boek nu!</button>
</h2>
```

In de class:

```
// Order plaatsen. Event emitten voor deze stad.
// Dit gaan opvangen in city.orders.ts
order(city) {
   console.log(`Stedentripje geboekt voor: ${this.city.name});
   this.orderService.Stream.next(city);
}
```

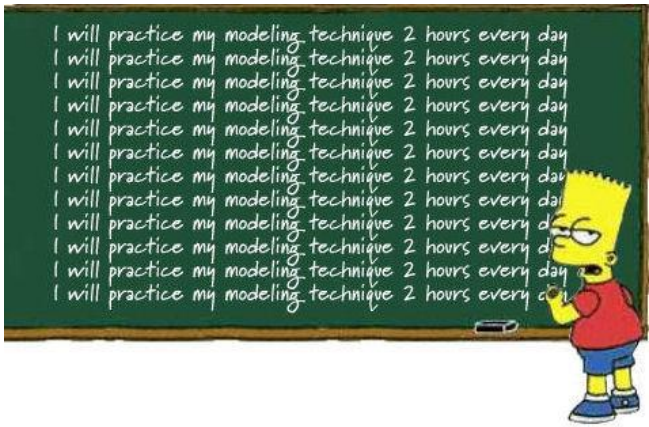
3. Subscriber component

```
//city.orders.ts - Een soort 'winkelmandje',
// bijhouden welke stedentripjes zijn geboekt.
import …

@Component({
   selector: 'city-orders',
   template: `
   <div *ngIf="currentOrders.length > 0">
      …
})

export class CityOrders {
   …
   ngOnInit() {
      this.orderService.Stream
         .subscribe(
            (city:City) => this.processOrder(city),
            (err)=>console.log('Error bij verwerken City-order'),
            ()=>console.log('Complete...')
         )
   }
`    …
}
```

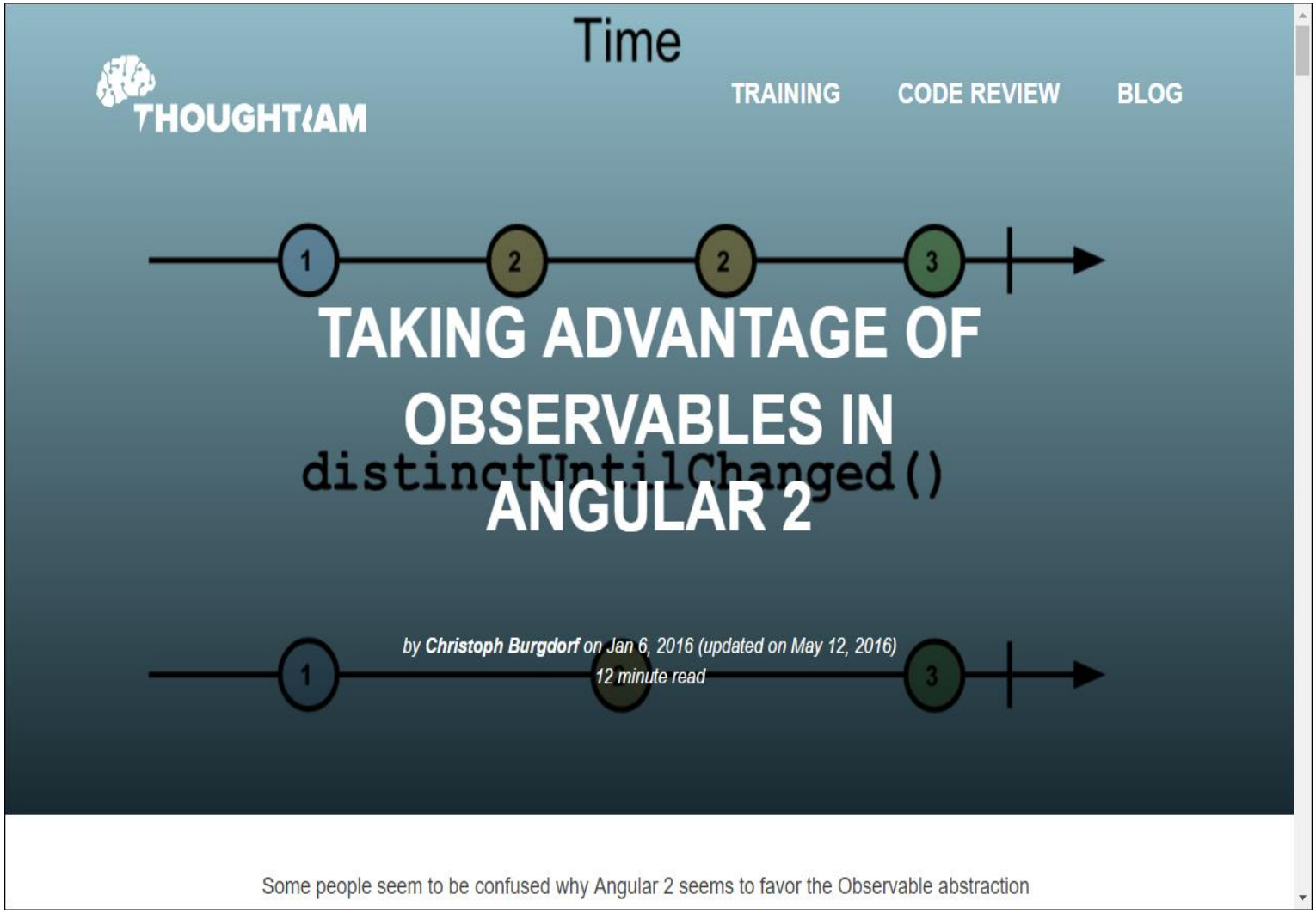
Checkpoint

- Event Bus : ‘onzichtbaar’ werken met Streams en `Subject`
- Er zijn opties voor het werken met Observable Streams.
- Voorbeeld: `\303-pubsub-ordercomponent`
- Oefening `6e)` e-commerce applicatie maken.

Oefening….



Meer over Observables



http://blog.thoughtram.io/angular/2016/01/06/taking-advantage-of-observables-in-angular2.html

Andre Stalz – introductory article

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The introduction to Reactive Programming you've been missing

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The introduction to Reactive Programming you've been missing

(by @andrestaltz)

This tutorial as a series of videos

If you prefer to watch video tutorials with live-coding, then check out this series I recorded with the same contents as in this article: Egghead.io - Introduction to Reactive Programming.

CORY RYLAN WORK CONTACT

My name is Cory Rylan, Senior Front End Engineer at Vintage Software and Angular Boot Camp instructor. I specialize in creating fast, scalable, and responsive web applications.

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Angular 2 Observable Data Services

Nov 17, 2015
Updated May 6, 2016 - 8 min read

Angular 2 brings many new concepts that can can improve our JavaScript applications. The first new concept to Angular is the use of Observables. Observables are a proposed feature for ES2016 (ES7). I wont go in depth into Observables but will just cover some of the high level concepts. If you want a introduction to Observables check out my screen cast.

INTRO TO RXJS OBSERVABLES AND ANGULAR 2

The rest of this post will cover more data and application state management in a Angular 2 application. At the time of this writing Angular is on version Beta 1. This post has been updated as of Beta 15. The syntax of how Observables and their

https://coryrylan.com/blog/angular-2-observable-data-services

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Observables In Angular 2.0

Author: Torgeir Helgevold

Published: Wed Jan 06 2016

Viewed 3375 times

The RxJs community has presented the idea that any series of events can be modeled as one or many asynchronous or synchronous arrays. In the following post I want to explore this by modeling a series of different user inputs as Observables.

I am still learning about Observables and their potential, but I figured it would be interesting to implement a custom text editor, from scratch, using Observables to represent keyboard and mouse events.

Building a perfect text editor is not really the point here, but I want to see if there is any added value from looking at input sequences as Observables. The first step when building a text editor is identifying which input events to support. In my sample I have decided to focus on adding the ability to input and delete characters. Currently I have limited the input

http://www.syntaxsuccess.com/viewarticle/observables-in-angular-2.0